

Gene Expression Project: Clustering and Deep Learning of Single-Cell RNA-Sequencing Data for Mouse Brains

Nikita Jayaprakash

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1 Introduction

The objective of this analysis is data exploration and cell-wise clustering of single-cell RNA sequencing data for mouse brains. Different methods of unsupervised learning, including K-means, principal component analysis, Poisson mixture models, and variational autoencoders, were used to see how the analysis changes with varying complexity and statistical techniques.

This analysis aims to answer the following questions:

- Regarding clustering, can the 3005 cells be partitioned into biologically meaningful groups that align with known cell types?
- Regarding visualization, can low-dimensional embeddings provide insight into the underlying structure and separability of the data?

This analysis concluded that the applications of principal component analysis (PCA) with K-Means clustering was the best method for partitioning the cells into biologically meaningful groups that aligned with the known cell types. While more complex methods were used, it is possible that they were capturing dynamics in the gene expression that were not relevant to the clustering of cell types since this analysis focused on unsupervised methods. For visualization, the t-distributed Stochastic Neighbor Embedding (t-SNE) method was the most clear in terms of showing different clusters after applying a clustering method, but it did not help in reducing the noise for clustering with KMeans.

1.1 Data

The dataset used in this analysis was originally used by Zeisel et al.'s "Cell types in the mouse cortex and hippocampus revealed by single-cell rna-seq" paper [2]. The data includes gene expression for thousands of genes. Since this dataset is solely about gene expression, there is no relevance to ancestry. In addition, it includes the labels of seven cell types present in the data: interneurons,

pyramidal SS, pyramidal CA1, oligodendrocytes, microglia, endothelial-mural, and astrocytes ependymal. Since this analysis is an exploration of unsupervised learning, the labels are not used during the training process. They are used to assess the accuracy of different methods by determining if the method can recover these groupings without using the labels.

2 Methods

This analysis explores different methods of dimension reduction, both for data visualization and improved clustering by reducing noise in the data, and unsupervised clustering methods of varying complexity in order to recover the original cell types. First, exploratory data analysis was used to determine the complexity of the data. Due to the skewed nature of the counts, all counts were also log-transformed.

This analysis began with a naive method of simple K-means clustering on the raw data without outlier removal or extensive gene filtering. The poor performance of this metric due to excessive noise in the data highlighted the need for dimension reduction.

First, the dimensions were reduced fairly manually by isolating the top 80 most-variable genes since most of the genes have limited expression in this dataset. Then, after isolating the most-variable genes, this analysis compared K-means when applied to principal component analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE). These two methods of dimension reduction were compared for how well they reduced dimensions in a way that helped the following clustering methods.

In all applications of K-means so far, there was consistent overlap in some of the clusters. This highlighted a potential need for soft probabilistic clustering. This led to an application of multivariate Poisson mixture models with expectation-maximization step, which was very superficially inspired by Meizia Ye et al.’s paper, "A multi-Poisson dynamic mixture model to cluster development patterns of gene expression by RNA-seq" [1]. Since Poisson distributions are used to model counts data, it was the ideal distribution for this mixture model, especially since the gene expression is so skewed that it likely does not approximate a Gaussian distribution. Note that this method used the raw counts rather than the log-transformed counts to maintain model assumptions.

Finally, this analysis used variational autoencoders (VAEs) because the increased complexity of deep learning techniques could potentially capture more complex behaviors in clustering the cell types. This analysis used two different types of activation functions (ReLU and sigmoid) and experimented with one to three hidden layers.

When comparing the clustering accuracy of these unsupervised clustering techniques, this analysis used Adjusted Rand Index (ARI), which is a chance-adjusted similarity measure for clustering results, and Normalized Mutual Information (NMI), which quantifies the agreement between two partitions. Larger values closed to 1.0 in both metrics show perfect correlation between two cluster

assignments.

3 Results

3.1 Exploratory Data Analysis

This dataset includes 3005 cells with 1005 genes each. As shown in Figure 1, there are differences in the variation of gene expression across the different cells and the different genes. Because of this, this analysis determined the top 80 most variable genes and used them as a focus for majority of the techniques. These most-variable genes are highlighted in red in the right plot in Figure 1.

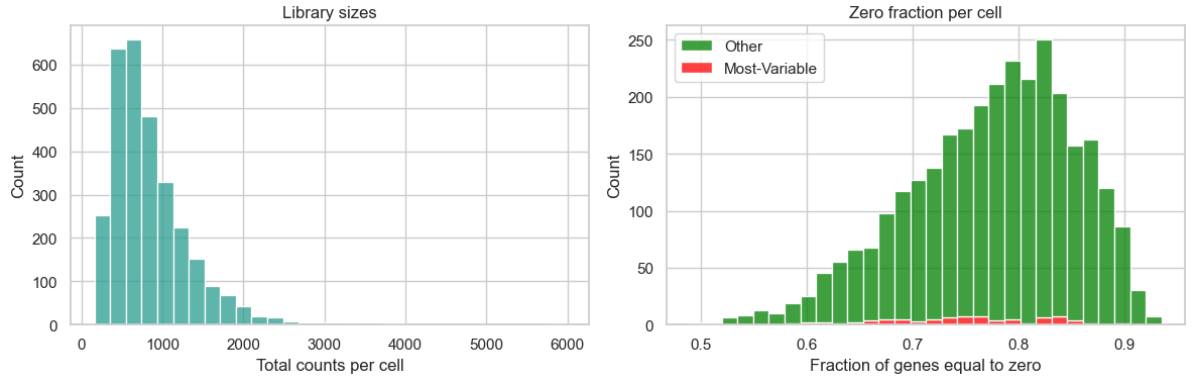


Figure 1: The total counts of genes per cells is strongly right-skewed, and the majority of genes are not expressed in over 50% of the cells.

3.2 Model Performance

3.2.1 Overall Comparison

Based on Table 1, the relative ordering of the different clustering performance (from best to worst) is the same across the two different metrics, ARI and NMI. Overall, the strongest clustering performance is from PCA(8) with K-Means, followed closely by t-SNE with K-Means. For the applications of VAE, this analysis used the ReLU and the sigmoid activation function and compared 1 to 3 hidden layers for each activation function. Among the VAEs with the ReLU activation function, the 2-layer 2D VAE latent mean had the strongest clustering performance. Among the VAEs with the sigmoid activation function, the 1-layer 2D VAE latent mean had the strongest clustering performance. For all VAE models, the 1-layer 2D VAE with the sigmoid activation function had the best performance, although it was still worse than PCA or t-SNE with k-Means. Poisson mixture models did not have strong clustering performance and was only better than the naive K-means method.

Representation	ARI	NMI
Naive KMeans	0.196	0.353
PCA(8) + KMeans	0.796	0.748
Poisson Mixture Model	0.367	0.495
t-SNE + KMeans	0.769	0.742
1-Layer 2D VAE latent mean (ReLU) + KMeans	0.441	0.492
2-Layer 2D VAE latent mean (ReLU) + KMeans	0.659	0.667
3-Layer 2D VAE latent mean (ReLU) + KMeans	0.569	0.628
1-Layer 2D VAE latent mean (Sigmoid) + KMeans	0.694	0.667
2-Layer 2D VAE latent mean (Sigmoid) + KMeans	0.548	0.618
3-Layer 2D VAE latent mean (Sigmoid) + KMeans	0.507	0.642

Table 1: PCA(8) with K-Means had the best performance in recovering the clusters from gene expression, followed by t-SNE(2) with K-Means.

3.2.2 Dimension Reduction and Clustering

We began with a naive approach of using K-means on the raw counts data of gene expression. As shown in Figure 2, this method fails to recover majority of the cell types and requires significant improvement through noise reduction.

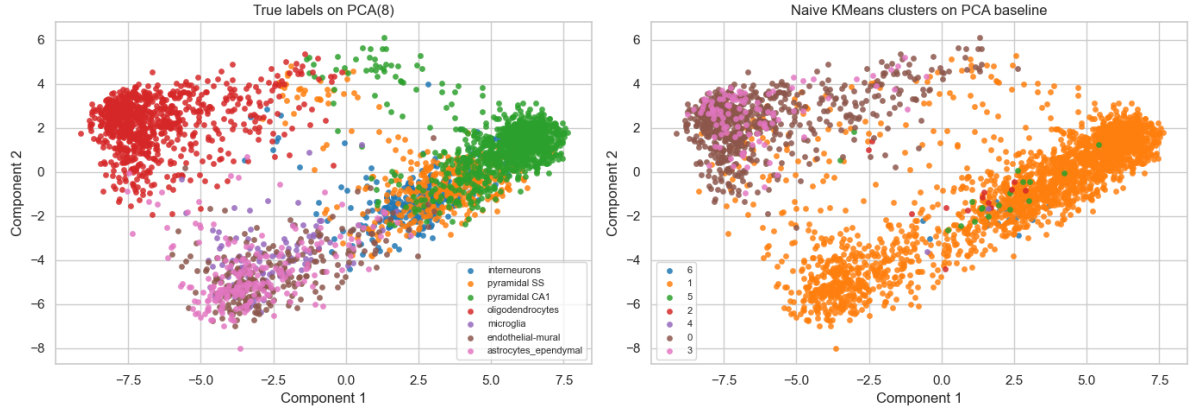


Figure 2: The naive K-means uses the raw data for clustering, but the first two principal components are used to visualize these naive clusterings.

After isolating the top 80 most-variable genes, this analysis uses principal component analysis to determine variables that maximize the variability in these genes. Based on the screeplot in Figure 9, the elbow occurs at $k = 8$, so this analysis will use the top 8 principal components when these scores are used for k-Means. As shown in 3, this strongly improves the clustering and the recovery of the original cell types.

This analysis also used t-SNE for dimension reduction. While PCA is a

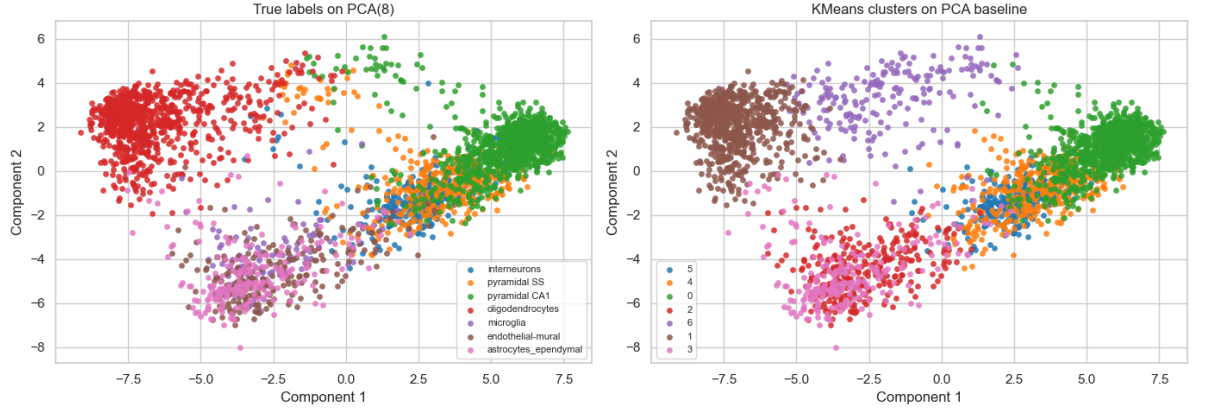


Figure 3: Running K-means on the principal components drastically improves the clustering, but there is still overlaps in the predicted clusters and the true labels.

widely used method for dimension reduction, t-SNE is more often used in applications of gene expression. As shown by Figure 4, this is an improved visualization technique for visualizing these clusters. As previously mentioned, it does not improve the clustering performance of k-means when compared to PCA.

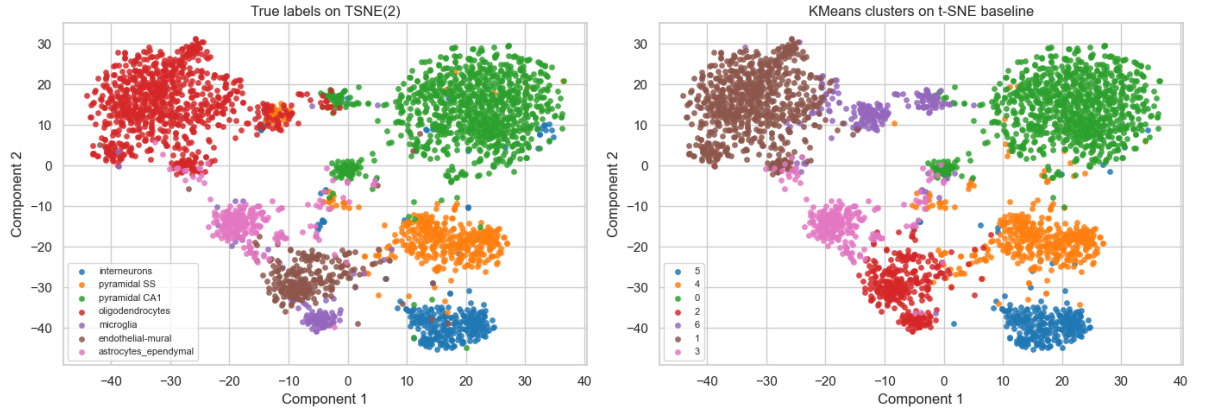


Figure 4: In comparison to PCA; t-SNE is better at visualizing clusters but fails to recover the true cell-type labels.

Since both the clustering of PCA and t-SNE had overlaps in the clusters, this analysis transitioned to Poisson mixture models in order to provide soft probabilistic clustering. As shown by Figure 5, this method seems to over-correct. It finds too many overlaps in the true cell labels and struggles to find definitive clusters.

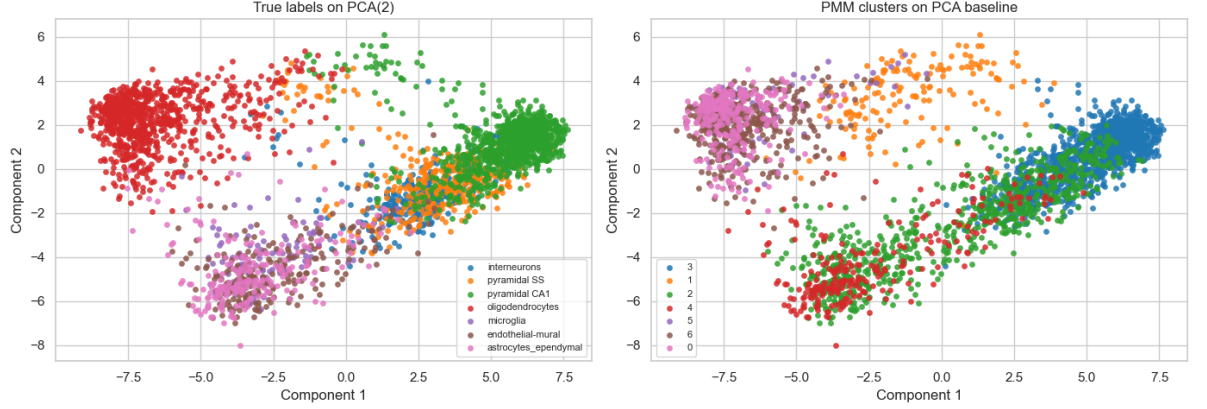


Figure 5: Similar to the naive K-means approach, the Poisson mixture models uses the raw data for clustering, but the first two principal components are used to visualize these clusters.

3.2.3 Deep Learning Extension: Variational Autoencoders

The performance of the 2-layer 2D VAE model with the sigmoid activation function, as shown in Figure 6, seems to have a similar layout of the latent space to the PCA with k-means approach: the clusters seem to be similar to the larger cluster labels, but it fails in identifying the small clusters and the cells that exist between the larger clusters in the latent space.

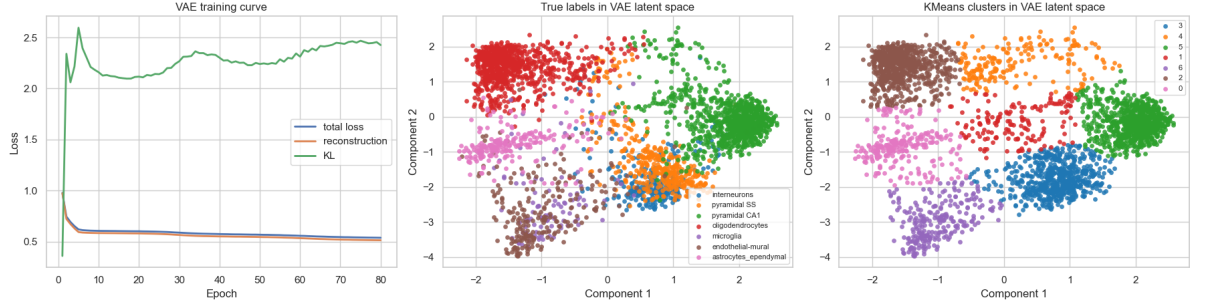


Figure 6: Of the VAE models with the ReLU activation function, the best clustering was from the 2-layer 2d VAE model.

There are very clear separable clusters in the 1-layer 2D VAE with sigmoid activation latent space, as shown in 7. However, it is not as separable as the t-SNE clusters, and it fails to recover the cell types in its clustering. This shows that it is creating strong separation but not in regards to the cell types.

In terms of latent space clustering, the results of clustering performed directly on the low-dimensional latent embeddings show strong clustering, espe-

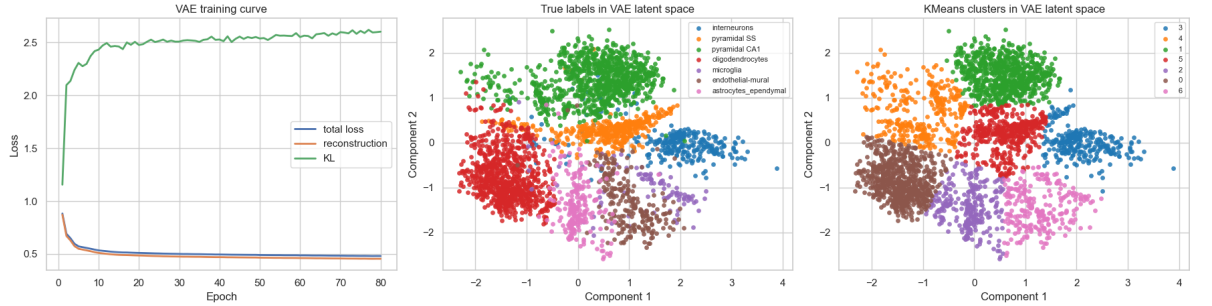


Figure 7: For the VAE models, the 1-layer 2D VAE with sigmoid activation function had the best clustering.

cially for the larger clusters, for both VAE latent spaces shown above. However, it fails to identify the smaller clusters that are spatially closer in the center of this latent space, which could explain the stronger performance by K-means on PCA and t-SNE.

4 Discussion

4.1 Dimension Reduction and Clustering Techniques

Overall, among the unsupervised clustering techniques, K-means performs better with clustering than Poisson mixture models. K-means with PCA was not outperformed by the increased complexity of the VAE techniques. Among the two dimension reduction techniques, t-SNE method with K-means had the best visualization, but PCA performs better with clustering.

4.2 Deep Learning Extension: Variational Autoencoders

In terms of architecture and training, the hidden dimensions of each layer was maintained at 64 dimensions for all VAE models in order to maintain complexity. Experimenting with higher dimensions worsened the performance of the model, likely due to high dimensionality with only a sample size of 3005 cells. This analysis used to different types of activation functions, ReLU and sigmoid, to see if either performs better with gene expression data. For each activation function, this analysis fit 1 to 3 hidden layers in order to see which performed the best and when added complexity worsened the performance. For representation justification, the input representation was the top 80 most-variable genes in order to use relevant genes. There was no additional dimension reduction used because this analysis still wanted to use a sizable amount of data to train these complex models. Using further dimension reduction would reduce the amount of data that could be inputted.

In addition, while experimenting with these different architecture, the 3-hidden-layer VAE with the sigmoid activation function creates a strange shape in the latent space, as shown in Figure 8. While it does create clear clusters across its curves, it fails to separate some of the larger clusters. This also shows that added complexity can create interesting, but ultimately unhelpful, structures in the latent space.

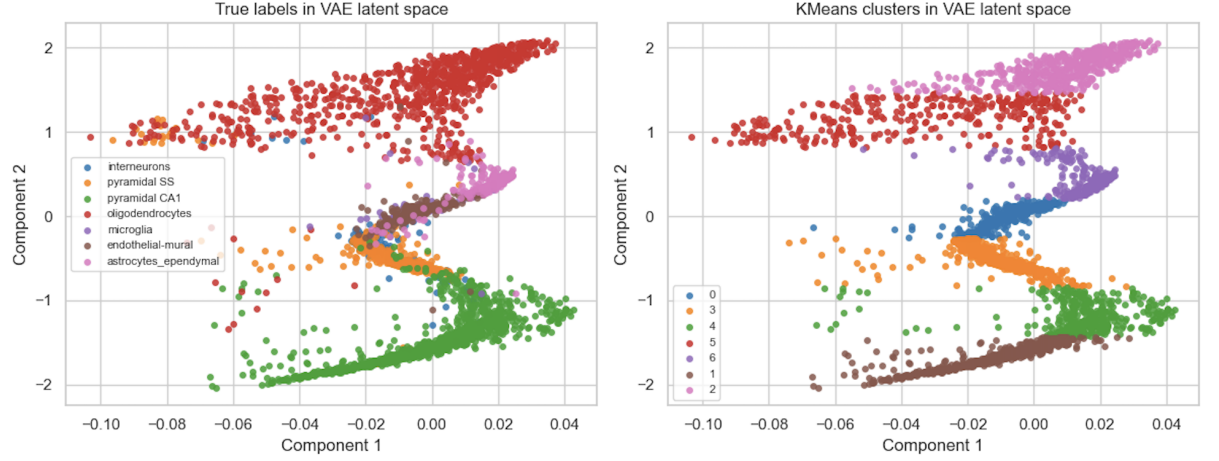


Figure 8: Your descriptive caption goes here.

For a direct comparison between the performance of this deep learning pipeline and the "classical" pipeline, the visualization of t-SNE has better separation and hence most visually-intuitive clustering than the VAE clustering, and the clustering performance is stronger with PCA as the input of K-means rather than VAE. This could be due to the added complexity of VAE models capturing complex behaviors that do not directly correlate with the true clustering labels. Since this analysis solely used unsupervised clustering techniques, there is no guarantee that the predicted clusters are separated by the cell types, as shown by the poorer performance of the deep learning techniques.

5 Appendix

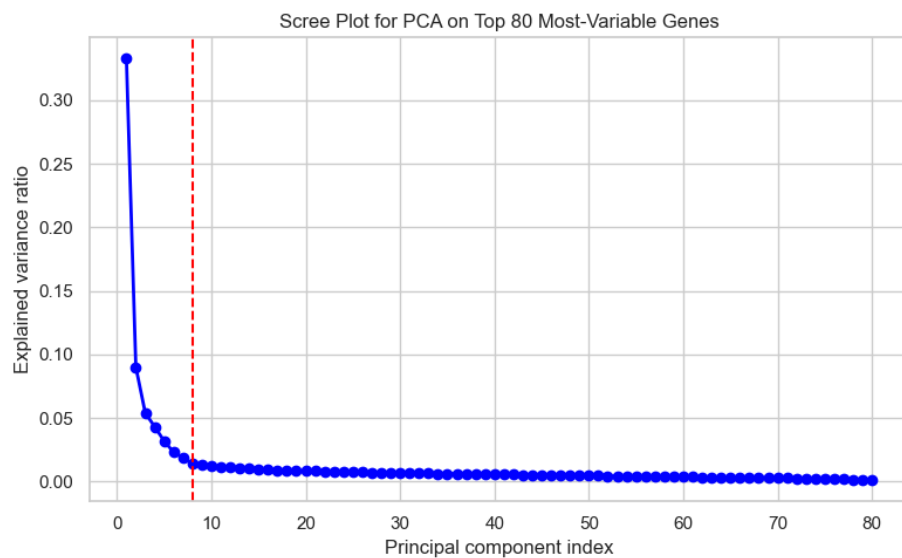


Figure 9: Your descriptive caption goes here.

References

- [1] Meixia Ye et al. “A multi-Poisson dynamic mixture model to cluster developmental patterns of gene expression by RNA-seq”. In: *Briefings in Bioinformatics* 16.2 (2015), pp. 205–215. DOI: 10.1093/bib/bbu013. URL: <https://academic.oup.com/bib/article/16/2/205/246602>.
- [2] A. Zeisel et al. “Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq”. In: *Science* 347.6226 (2015), pp. 1138–1142. DOI: 10.1126/science.aaa1934.